Spontaneity: 1) $(V_1 \text{ vs. SHE} - V_2 \text{ vs. SHE}) > 0$ } This will give a spontaneous (galvanic cell)
If < 0, take $V_2 - V_1$ }

Rxn @ more negative potential proceeds in oxidation direction

→ The more negative the potential, the more likely oxidation is to occur.
→ The no. of e^- involved does not matter to determine V_{cell} (since they are all normalized by the SHE)

Nernst Corrosion + Pourbaix diagrams
↳ Tell us about stability/passivation of metal electrodes as a function of potential + pH

+ Constructed w/ Nernst eq.
+ Tell us how susceptible metals are to corrosion via H_2 or O_2

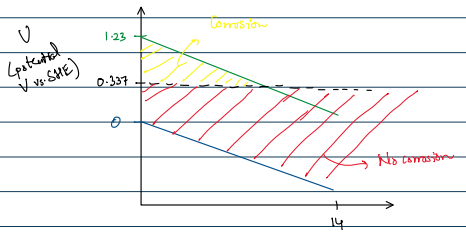
eg: $P_{H_2} = 1$ bar, $P_{O_2} = 1$ bar ; V_{H_2/H^+} & V_{O_2/H_2O} as function of pH

$H^+ + 2e^- \rightarrow \frac{1}{2} H_2$

$(V_{H_2/H^+} - V^\circ_{H_2/H^+}) = (V^\circ_{H_2/H^+} - V^\circ_{H_2/H^+}) - \frac{RT}{F} \ln\left(\frac{P_{H_2}^{1/2}}{C_{H^+}}\right)$ ($V^\circ_{H_2/H^+} = 0$)

$V_{H_2/H^+} = -0.0591 \times pH$ V vs. SHE

Similarly,
$V_{O_2/H_2O} = 1.23 - 0.0591\ pH$ V vs. SHE



Will a metal corrode?

$Cu \rightarrow Cu^{2+} + 2e^-$; $V_{Cu/Cu^{2+}} = \left(V^\circ_{Cu/Cu^{2+}\ vs.\ SHE}\right) - \frac{RT}{2F} \ln C_{Cu^{2+}}^{-1}$

$= 0.337 + \frac{RT}{F} \ln C_{Cu^{2+}}^{1/2}$ ← Not dependent on pH

Stability: • $(V_{Cu/Cu^{2+}} - V_{H_2/H^+}) > 0$ Spont ✓

↑ This one is more negative (i.e. this is oxidized)

∴ Cu does not corrode

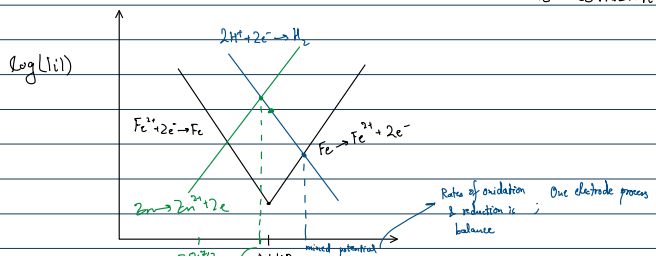
• $(V_{O_2/H_2O} - V_{Cu/Cu^{2+}}) > 0$ Spont ✓

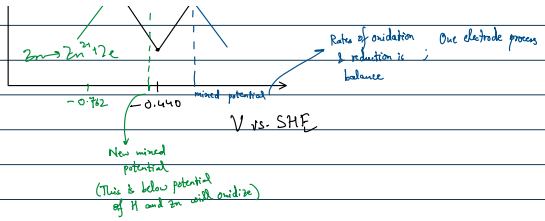
↓ More neg (oxidation)

∴ Cu does corrode

Current density vs. potential

What is corrosion potential & corrosion rate?





Electrochemical Transport

(1) Species can move via convection, diffusion and electrical migration

$$\bar{N}_j = \underbrace{-z_j u_j F c_j \nabla\phi}_{\text{migration}} \underbrace{- D_j \nabla c_j}_{\text{Diffusion}} + \underbrace{c_j \bar{v}}_{\text{Convection}}$$

u_j: Ionic mobility; $\nabla\phi$: gradient of potential; $\bar{v}$: fluid velocity (Navier Stokes)

↳ migration: how do things move in response to electric field ($-\nabla\phi$)
- ϕ represents electrostatic potential
- u_j describes avg. velocity of species acted upon by 1 N/mol force
- $z_j F$ → charge per mole of species

$\therefore \underbrace{(u_j F z_j \nabla\phi)}_{cm/s} \cdot \underbrace{c_j}_{mol/cm^3}$; for neutral speis: $z_j = 0$

↳ Diffusion: how do things move in response to conc. driving force?

↳ Convection: how do things move in response to bulk motion of medium

Valid for dilute solutions → D_j constant and equal D_j of j @ infinite dilution
↳ No frictional forces on j besides solvent (molten salt)

(2) Flow of a charged species constitutes an electric current

↳ Current in electrolyte is due to motion (flux) of charged particles

$\bar{i} = F \sum_j z_j \bar{N}_j$; $\bar{i}$ is a vector quantity

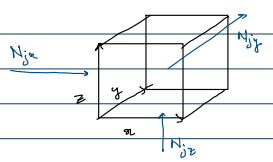
(3) Mass is conserved

$Acc = In - out + gen - cons.$

$$\frac{dc_j}{dt} = -\nabla \cdot \bar{N}_j + R_j$$

↳ generation in homogenous solns (not electrode process) i.e. $R_j = 0$ } No reaction in the bulk

electrode reactions will be BC



In 1D: $\lim_{\Delta x \to 0} \frac{N_{jx}|_x - N_{jx}|_{x+\Delta x}}{\Delta x} = -\frac{\partial N_j}{\partial x}$

(4) Electroneutrality is imposed everywhere

$$\sum_i z_j c_j = 0$$

These principles enable us to describe current flow in electrolyte